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Public Interest, Private Profit: Regulating Germline Editing of Human Genes

Introduction: The Interpretive Puzzle

“Cheaper, Faster, Riskier”: this is how one source describes a new gene-editing technology that is becoming a reality—namely, CRISPR/Cas9—and poised to redefine the concept of "precision medicine” (Furrow, 33). This is an “unprecedented" rate of innovation that is clouded by “profound ethical and economic worries—and they have to do with issues of fairness in biotech and life-prolonging biologies" that in this instance include those of “life-saving biologies" (Courtright-Lim & Drago, 444). In essence, of course, it is a question of property and responsibility. “Modernity's bioscience" is based upon “the giants of public investment—that is, research cultures supported by public research grants and public data inputs" that are threatened with a new reality of “monopoly control by private firms", or as explained by the term of ‘Proprietary Enclosure,’ which is explained later in this paper (Capps et al. 20). This reality is a danger to increased inequities in healthcare delivery. The research question to study and analyze through the information brought by these sources is: "Should gene editing tools such as CRISPR-Cas9 be regulated through a public system based upon the principle of genomic solidarity to require fair distribution and prevent the monopoly control of technology that grew out of a huge public investment?"

This paper will seek to show that the failure of governance mechanisms to take into account the public provenance of genomic technology has led to a framework of "catallactic bias" that is market-oriented and which prioritizes profit at the expense of universal access. In contrast, an ethical framework such as that of Genomic Solidarity is needed—a history-grounded critique that mandates public sharing of benefits as well as public responsibility of gene editing technology— and is clearly the best possible foundation upon which a regulatory system could be developed in order to moderate its social risks.

The Conflict of Public Investment and Proprietary Enclosure

The core problem of Germline Genome Editing (GGE) is not purely one of technology but of property and responsibility. The science underlying this field, including the CRISPR/Cas9 technology, did not emerge in a private vacuum; it is rooted in "the giants of public investment" (Capps et al. 20). This extensive public backing includes "research cultures supported by public research grants and public data inputs," all of which constitute a non-negotiable claim to a public interest in accountability (Capps et al. 20). However, this foundation is currently being threatened with a new reality of "monopoly control by private firms," which is a danger to increased inequities in healthcare delivery (Capps et al. 20). This structural tension establishes the framework for the ethical analysis: the public's investment—its moral ledger—is being systematically converted into private, proprietary wealth, creating the condition that Capps et al. term "catallactic bias."

The tension between the public origins of genomic science and its private destination defines the ethical landscape of CRISPR. Capps, Chadwick, Joly, and others articulate this conflict, stating that “the contemporary 'biotech industry is dominated by a 'catallactic bias': that is, a systemic bias preferring 'private', 'commercial', and venture-driven 'R&D', and this will frequently entail proprietary appropriation of a 'global common' science and biotech resource” (Capps et al. 20). Since this is a market-driven sector, the resulting failure is that “the cost of biotech and life-prolonging biologies is increasingly prohibitive” (Courtright-Lim & Drago 448). The entire dilemma surrounding the CRISPR-Cas9 tool is encapsulated by the extended metaphor that Furrow describes: “Cheaper, Faster, Riskier” (Furrow 33). The "Cheaper, Faster" components describe the breakthrough's technical efficiency and its financial appeal to the private sector, reducing time and cost barriers and opening the floodgates for mass commercial deployment. Conversely, the "Riskier" component describes the consequence of that speed and commercial aggression, meaning that the technology has advanced far beyond the existing legal and ethical frameworks necessary to govern it.

This framing establishes publicly provided infrastructure and open science standards as a continued, non-revoking “moral mortgage” owed back to society at large. Capps et al. continue to explain that “innovations in science and technology—a basic biological foundation, as well as a basic infrastructure of clinical trials—have created a global common” that is immediately “commercially enclosed through patent and data ownership,” a phenomenon that Capps et al. define as “catallactic bias” (Capps et al. 20). This highly descriptive term outlines a system-level preference based upon a commitment to market value above public need that leads revolutionary discoveries such as CRISPR from public use to private gain. The ultimate failure is that this

government funding, which built the very foundation for CRISPR's success, is what private research now uses to justify its immediate, proprietary enclosure of the resulting science for market gain.

The term "the giants of public investment" is a metaphor used to establish a "moral ledger" owed to society, representing more than just basic financial backing (Capps et al. 21). This investment directly refers to the "research cultures supported by public research grants and public data inputs," which constitute the public's moral claim (Capps et al. 20). The core conflict is that this public foundation is immediately subject to proprietary enclosure, defined as keeping scientific research and information from public view so private companies can profit. This immense public investment is being routinely transformed into private wealth under the threat of "monopoly control by private firms" (Capps et al. 20). This structural flaw, the "catallactic bias," makes the cost of biologies "increasingly prohibitive," actively endangering health equity (Courtright-Lim & Drago 448).

The apparent economic tension—where a technology is described as "cheaper, faster, riskier" (Furrow 33)—is inextricable from the qualities inherent in CRISPR. The RISK component profoundly affects market dynamics, leading to highly expensive final products because the potential for "uncertain risks" at a population-level assessment (Furrow 51) and inherent legal uncertainty necessitate massive capital investment to mitigate liabilities. This pursuit of "faster" innovation, driven by profit, raises development costs because private enterprises must embed the cost of future systemic or regulatory risks directly into the final therapeutic price. This makes equitable access economically prohibitive, despite the tool being

technically easier and cheaper to use. The “cheaper” and “faster” qualities of CRISPR make it a completely suitable research tool for market-directed R&D because they enable private enterprises to proceed at a rate of innovation that exceeds that of traditionally considered ethical and regulatory standards. This market demand for “faster” innovation induces a regulatory gap, as the application rate exceeds the law’s ability to properly understand its profound implications and manage those “uncertain risks” (Furrow 51). Ultimately, the core problem is the interaction of an efficient “tool” and a “highly aggressive profit incentive” (Furrow 51). This focus ensures the technology is funneled toward private monopolies, preventing public access.

My analysis reveals that the fundamental threat posed by Germline Genome Editing (GGE) is not the risk inherent in the technology itself, but the systemic risk created by the interaction between the technology's efficiency and the current governance structure. Due to Furrow's observation that CRISPR is “riskier” because its application rate outpaces the law's ability to regulate it (creating “uncertain risks” at a population level), I believe this is precisely where Capps et al.'s economic critique must be applied. The speed of innovation is not a neutral development; it is a direct function of the “catallactic bias,” which prioritizes aggressive profit-seeking within commercially aggressive market sectors. This leads me to the conclusion that the “risk” is therefore a market-induced variable, not just a biological one, in terms of the risk of the technology itself. By relentlessly pursuing a “faster” pace of innovation, private firms use the foundational knowledge built by “the giants of public investment”—the very source that establishes the public's “moral ledger”—and intentionally create the regulatory gap that makes the entire enterprise “riskier” for society. Because of all of this, proprietary enclosure ensues, as the high costs associated with managing this systemic risk are then embedded into the final

product, rendering equitable access economically prohibitive for those who cannot afford it. Thus, my conclusion is that the ethical challenge is not one of slowing down science, but of structurally reforming the way it is governed. The market's pursuit of “faster” innovation is unable to work with the public's need for fair access unless the public's original investment is recognized and enforced as a right to benefit-sharing, where they also reap the benefits of the private research and technology, which can only be achieved through a principle like Genomic Solidarity.

Failure of Existing Legal Safeguards

The Ineffective Counterweight of March-In Rights

The proprietary enclosure driven by the “catallactic bias” (Capps et al., 20) persists because the primary legal safeguards intended to check it have been proven functionally inert. The Bayh-Dole Act formalized the ability of universities to commercialize federal research, thereby providing the structural legal foundation for the monopoly control the paper critiques. The Act contains a provision known as "march-in rights," which allows the government to cancel an exclusive patent license if a product is "not available to the American public on reasonable terms." This was intended as the critical legal counterweight to enforce public accountability against private profit. However, a qualitative study by Treasure, Avorn, and Kesselheim reveals that this mechanism is a "functionally ineffective counterweight" (761). The authors concluded that there is "little prospect NIH will ever invoke march-in rights" after documenting that, despite five public petitions since 1980, the National Institutes of Health (NIH) has denied every request (Treasure et al., 761). This evidence is crucial: it empirically illustrates that the systemic bias

described by Capps et al. (20) has translated into institutional and political reluctance, rendering the most powerful public protection mechanism entirely powerless against exorbitant pricing and monopoly control. The failure of "march-in rights" demonstrates that the current system lacks a practical legal remedy to uphold the public's "moral ledger" (Capps et al., 21).

In my view, the failure of the NIH to ever invoke march-in rights, as documented by Treasure, Avorn, and Kesselheim (761), is not due to a flaw in the law's wording, but a consequence of the "catallactic bias" being fundamentally institutionalized within the governance structure itself (Capps et al., 20). The NIH is politically paralyzed because utilizing this legal power would directly undermine the very commercial incentive system that the Bayh-Dole Act was designed to foster. If the government were to cancel a license due to high pricing, it would signal to the entire biotech sector that private investment—the source of the "faster" innovation the government structurally prioritizes—is vulnerable to public intervention. The institutional reluctance documented by Treasure et al. (761) therefore proves that the government places a higher political value on maintaining the "highly aggressive profit incentive" than on enforcing the public's "moral ledger" (Capps et al., 21). The market's structural dominance makes the cost of enforcing equitable access (by disrupting private monopolies) politically and economically too high. This leads to my conclusion: an internal, reactive legal mechanism like march-in rights is inherently incapable of checking an external, systemic economic bias. The tool intended to protect the public has become a dead letter precisely because the system is designed to favor the private wealth it was supposed to regulate.

Market Consolidation and Ethical Control

"The evident failure of such publicly funded checks" allows for private methods such as exclusive licensing to fully consolidate the marketplace and lead to sole-provider monopolies that control both entry and use (Richardson et al., 241-242). In this manner, an "investigation of genetic testing for Alzheimer's Disease" conducted by Skeehan et al. reveals this right of property at work and reveals that due to "exclusive licenses owned by Duke University, the marketplace for genetic testing has been consolidated into the sole provider, Athena Diagnostic Systems, Inc., such that it is now possible to speak of a `sole-provider consolidation of testing'" (Skeehan et al., S71). In such a manner, "the sole-provider consolidation of testing" makes it possible for "a single corporation to control the marketplace and better negotiate with insurers for coverage" and directly restrict access and go against the public good (Skeehan et al., S71-72). Most importantly for this analysis, of course, is that Skeehan et al.'s case study itself indicates that the sole-provider consolidation of testing is indeed possible due in part to the existence of an exclusive license that in some circumstances may be justified on grounds of profit and in other on grounds of enforcing professional medical standards and to prohibit premature and unguided testing (Skeehan et al., S71). However, this ethical justification ultimately fails because it reinforces the very proprietary enclosure that undermines public interest. The fact that the federal government, through the political failure of "march-in rights" (Treasure et al., 761), allows such monopolies to persist proves that the system is structurally incapable of ensuring both ethical standards and universal access outside of a mandatory, public-oriented framework.

In my assessment, the ethical justification offered by Skeehan et al. (S71)—that exclusive licensing can be justified to enforce professional standards or prevent unguided testing—is a political distraction that ultimately validates the “catallactic bias” (Capps et al., 20). While the intent may appear benevolent (preventing misuse), this justification is structurally flawed because it allows the very proprietary enclosure that creates economic exclusion to remain in place. The problem isn't that a single provider can set ethical standards, but that the system relies on a monopoly to do so. This creates a false choice between ethical control and equitable access. The evidence provided by the failure of "march-in rights" (Treasure et al., 761) demonstrates that the government lacks the political will to enforce public benefit-sharing. Therefore, the moment a sole-provider monopoly is granted, the primary incentive is always profit, as the ethical enforcement becomes a secondary, unregulated tool of the patent holder. Ultimately, the system is designed to favor private control, allowing the ethical facade to reinforce the structural injustice, thereby ensuring that the immense investment made by the “moral ledger” (Capps et al., 21) is perpetually converted into private, rather than public, benefit.

Genomic Solidarity: A Framework for Public Control for the Technology

The dialogue between the economic model proposed in Capps et al. and the legal/technical challenge proposed in Furrow, Treasure et al., and Skeehan et al. reveals how traditional ethical principles—such as individualized informed consent—prove inadequate to cope with this technology. Clearly, it is not merely an individual researcher's ethical decision or a firm's economic strategy that constitutes a challenge to society, but rather a failure of governance as a system that puts commercial profit above public security and universal accessibility as a

whole. For such a system to mitigate the "catallactic bias" as well as "uncertain risks" inherent within it, therefore, it is clearly imperative that a robust and realizable public norm—such as that of "Genomic Solidarity"—be introduced to secure public accessibility of public knowledge. Describing a component of Genomic Solidarity as defined by authors Mulvihill, Byers, and Wilfond, there is a proposed ethical imperative in sharing knowledge or benefits of genomic research (Mulvihill, Byers, & Wilfond, 19). Most pertinent to this discussion is its recognition of a crucial point of differentiation in terms of profit versus fair sharing. In particular, it draws on the precedent of the Bermuda Principles of the Human Genome Project, whose immediate public release of raw data averted its proprietary takeover at an early stage (Capps et al., 20). In this setting, a template in open-science research, a similar application of that precedent in CRISPR research through such mechanisms as grants that prevent public research from being obstructed or through placing basic patents developed from public research in a global common could effectively reshape an entire debate of regulatory oversight from a vague call of corporate social responsibility to a mandate of public right.

From all of this, I conclude that the necessity of a mandate like Genomic Solidarity comes from the realization that the ethical challenge posed by Germline Genome Editing is one of structural justice, not mere individual ethics. My prior analysis demonstrates that the "catallactic bias" (Capps et al., 20) is too deeply entrenched for internal legal checks, such as the "march-in rights" (Treasure et al., 761), to be effective. The problem isn't that we lack ethical guidelines, but that the system design allows the immense public foundation—the "giants of public investment"—to be leveraged to create a systemic risk (Furrow 51) that results in

economically prohibitive access. This systemic risk is the consequence of prioritizing the "faster" innovation of the market over public safety and accessibility. Therefore, placing the ethical burden on firms as "corporate social responsibility" fundamentally misdiagnoses the problem; the solution must be structural. Genomic Solidarity is the only principle that can work because it acts as a proactive mandate. Unlike reactive tools like march-in rights, a Genomic Solidarity mandate, modeled after the Bermuda Principles (Capps et al., 20), safeguards the "global common" by demanding benefit-sharing at the point of funding, preventing the proprietary enclosure from ever gaining structural dominance. This shifts the debate from pleading for access to enforcing a mandate of public right, ensuring the public's investment is finally realized as a public benefit.

Problem of Manageable vs. Societal Risk

(Counter-Argument)

An important opposite viewpoint that is commonly argued by those in favor of fast-tracked scientific research, such as in Rubeis & Steger's work, is that "the risks of Germline Genome Editing (GGE) could be managed in the long run" as they believe that individual biomedical risk could be minimized through further scientific research in the near term to tip the balance of benefits versus individual biomedical risks in their favor (Rubeis & Steger 133). In this sense, the "cheaper" and "faster" that come with CRISPR technology are a matter of strict ethical necessity in that it enables a fast-track discovery of curable measures for deadly single gene disorders and therefore that such lines drawn through a strict interpretation of a "first do no harm" or "principle of preventive action" in imposing biomedical research cannot be applied in a manner of endless delays in biomedical research to preserve human life. In this thinking, an

important overall specific social risk of engaging in biomedical research is seen as “manageable through an establishment of a regulatory framework before implementing a program of Germline Genome Editing" (Rubeis & Steger 134). Nevertheless, a crucial qualification is immediately provided by Rubeis and Steger when they state, “Since it is a matter of fact that it is 'extremely difficult to calculate adequately just how such modifications will affect a particular genepool,” “the medical risks involved for such a group may be too high” (Rubeis & Steger 134). This is a nod to the challenge that Furrow presents in terms of its legality and its ethical considerations, in that it is not merely an individual challenge but a social challenge. That is, it is a matter of not only determining how one may control or regulate an individual’s use of this technology, but also, through that technology’s use, influencing or impacting an entire society or social group in terms of its genotype or gene pool in a manner that is not fully understood at present. In reflecting on this tension between individual benefit and collective genepool risk, I interpret this not as a mere technical hurdle, but as the critical juncture where the “technological imperative” collides with our “moral ledger” (Capps et al. 21). While Rubeis and Steger suggest these risks are “manageable” through future regulation (Rubeis & Steger 134), my analysis suggests that the very market forces pushing for “faster” innovation—the “catallactic bias” (Capps et al. 20)—ensure that the regulatory framework remains perpetually behind the curve.

I have come to realize that the “faster” development Rubeis and Steger hold as an ethical necessity (Rubeis & Steger 133) is precisely what fuels the “regulatory gap” Furrow warns about (Furrow 51). By the time we fully understand the societal impact on the gene pool, the technology has already been enclosed by private patents, making access a privilege of the few

rather than a right for all. This realization moves my argument away from a simple debate over lab safety and toward a fundamental critique of how our economic structures weaponize scientific speed against public equity.

Conclusion

I began this paper with the research question of whether Germline Genome Editing (GGE) tools should be regulated through a public system based upon Genomic Solidarity to prevent the monopolization of technology grown from massive public investment. My analysis has shown that the ethical conflict is not an isolated incident of bad corporate behavior, but a structural failure of governance rooted in the “catallactic bias” (Capps et al.). This bias systematically converts the “moral ledger” established by the “giants of public investment” into private, proprietary wealth. This process is the direct cause of the “systemic risk” that I identified by synthesizing the work of Furrow and Capps et al.: the market’s relentless pursuit of “faster” innovation intentionally creates a regulatory gap, rendering the entire endeavor “riskier” and making equitable access economically prohibitive. Furthermore, the failure of internal legal checks—specifically the institutional paralysis of the NIH to ever invoke "march-in rights" (Treasure et al.)—confirmed my belief that the system is designed to favor the private wealth it was supposed to regulate. Even the ethical justifications offered for exclusive licensing by Skeehan et al. were revealed to be a distraction, at least politically, ultimately reinforcing the proprietary enclosure, undermining the public good.

The ultimate conclusion I reached is that the ethical challenge is not one of slowing down science, but of structurally reforming its governance. Relying on reactive legal tools is insufficient because the “catallactic bias” is systemic; therefore, the solution must be a proactive mandate. The Genomic Solidarity framework is the only viable path forward because, unlike "march-in rights," it compels benefit-sharing at the point of funding, guaranteeing that the public's investment is realized as a right to benefit. This shifts the focus from begging for access to enforcing a structural claim to justice. While this structural reform is necessary to ensure the tools of "precision medicine" fulfill their promise of universal benefit, its application raises new questions that require immediate, collaborative discussion. For instance, what specific financial and legal tools (such as non-obstruction clauses in grants) could the US government actually use to enforce "benefit-sharing" without totally nullifying patents, providing a necessary compromise between idealism and practical business realities? Additionally, how is the line between therapy and enhancement for germline editing explicitly managed by the laws currently in place in other highly developed countries, and what proof is there that their framework is actually "manageable" against the pressure of the aggressive profit incentive? Finally, how can a new regulatory body successfully oppose the biotech industry's political power to impose temporary restrictions on new, unknown risks, in light of the “technological imperative”? Addressing these questions is crucial for moving beyond simply identifying the structural problem and toward enacting a viable, publicly-oriented solution.

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